The Impact of Currency Dollarization Policy on Argentina's Economy

Litao Tan Jingyuan Xu Jiaxun Liu Jiayi Zhu

I. Introduction

Long before President Milei campaigned on *dollarization* as a key electoral platform in 2022, the concept had already been proposed by former Argentine President Carlos Saúl Menem as early as 1999. Menem's plan envisioned abandoning the dollar-pegged exchange rate regime then in place, advocating instead for the complete replacement of the Argentine peso with the US dollar.

There are two forms of dollarization: *de jure* and *de facto*.

De jure dollarization means officially adopting a foreign currency as legal tender, replacing the local currency in daily economic functions.

De facto dollarization happens when the foreign currency is widely used without legal status, often alongside a fixed exchange rate.

Argentina’s experience with de facto dollarization can be traced back to the 1990s. After facing hyperinflation in the late 1980s, the government introduced the Convertibility Plan in 1991, fixing the peso to the U.S. dollar at a 1:1 rate. This reform, combined with financial liberalization and tighter fiscal policies, initially restored economic stability. Inflation plummeted, growth returned, and foreign investment increased. However, over time, fiscal deficits widened, unemployment grew, and foreign reserves fell. In 2002, Argentina abandoned the fixed exchange rate, and the peso sharply depreciated. Over the next two decades, the peso continued to lose value, with the exchange rate increasing more than fiftyfold by 2023. Meanwhile, Argentina struggled with persistent fiscal deficits, high inflation, and growing debt, made worse by the COVID-19 pandemic and the Federal

Reserve's interest rate hikes. These pressures severely weakened the peso and deepened the economic crisis. In response, Milei proposed a radical "shock therapy" policy centered on de jure dollarization. His idea was not without precedent: Ecuador successfully dollarized in 2000, achieving low inflation, reduced country risk, better access to international credit, and a reversal of capital flight.

Given Argentina's ongoing challenges and the mixed lessons from Latin American countries, our study is trying to figure out the the prospects and potential outcomes of Argentina's dollarization policy.

II. Literature Review

The impact of currency dollarization policies has long been an important topic in the study of the economies of Latin American countries. Mcdermott (2024) analyzes the monetary policy histories of Ecuador, El Salvador, Liberia, Panama, and Zimbabwe and argues that while dollarization can provide short-term relief from inflation crises, its long-term economic benefits are vulnerable to the high-spending policies of populist governments.[1] Therefore, for dollarization to be successful, it must be accompanied by constraints on executive power and independent judicial oversight of the state's access to reserves. Similarly, Curutchet (2001) examines the potential costs and benefits of Argentina's dollarization in 1999 and finds that it could attract greater investment from the United States, reduce exchange rate risk costs and real interest rate volatility, and enhance financial stability.[2] However, he also highlights a major drawback: Argentina would lose its

[1] McDermott, Dante, "A Dollar a Day Keeps Inflationary Crisis Away: An Evaluation of Proposed Dollarization in Argentina" (2024). CMC Senior Theses. 3637. https://scholarship.claremont.edu/cmc_theses/3637

[2] Alfredo Schclarek Curutchet, "The Benefits and Costs of Official Dollarization for Argentina" (2001), Department of Economics, Lund University

"exit option," meaning it would no longer have the ability to implement independent monetary policy in the future.

Building on the importance of institutional factors, Spruk (2019) traces Argentina's inflation history and argues that institutional stability is key to long-term inflation control.[3] He cautions that without accompanying institutional reforms, dollarization may fail to curb inflation effectively and would not fundamentally resolve the country's economic instability. Furthermore, Wieber (2024) provides empirical evidence that dollarization can help reduce inflation, particularly in economies suffering from currency depreciation and monetary instability.[4] Nevertheless, he warns that Argentina's historically high economic volatility makes the loss of monetary sovereignty a significant risk, as it could leave the country without policy flexibility in the face of external shocks, potentially worsening economic instability.

Although these articles provide historical experiences, institutional requirements, and theoretical frameworks for dollarization, they are insufficient for directly analyzing the specific impact of Argentina's new dollarization policy since 2023. Studying dollarization after 2023 requires a focus on the latest economic data, the global financial market environment, Argentina's political landscape, and social impacts, while also incorporating contemporary case studies for a more comprehensive analysis.

III. Theoretical Framework

This paper focuses on three key variables: the real interest rate (R), the country risk (Risk), and the dollarization index (DI).These indicators capture the critical channels through

[3] Rok Spruk, "The rise and fall of Argentina", Latin American Economic Review, (2019) 28:16.
[4] Wieber, Chloe, "Dollarization in Latin America" (2024). Honors Theses. 3856. https://scholarworks.wmich.edu/honors_theses/3856

which a transition to a dollarized monetary regime may affect inflation dynamics, investor confidence, and overall financial conditions.

The real interest rate is defined as:

$$R = I - \pi$$

where we denotes the nominal interest rate and π represents the inflation rate. R reflects the inflation-adjusted return on financial assets and serves as a proxy for the real cost of borrowing and the incentive to save. In persistently high-inflation environments, nominal rates may be elevated, yet real returns often remain negative, discouraging saving and investment.

The country risk is proxied by the spread between domestic and U.S. sovereign bond yields:

$$Risk = I - I_{USA}$$

where I and I_{USA} are the nominal interest rates of Argentine and U.S. government bonds of similar maturity(In practice, this spread is typically measured using three-month government bond yields). This spread reflects perceived sovereign risk and is widely used as an indicator of a country's macroeconomic and fiscal credibility.

The degree of financial dollarization is measured by the share of foreign currency deposits in the broad money aggregate:

$$DI = \frac{FCD}{M2}$$

where FCD denotes foreign currency-denominated deposits and M2 includes local currency in circulation, demand and time deposits, and foreign currency deposits. This measure captures de facto dollarization in the banking sector.

We consider two principal transmission channels through which dollarization may influence macroeconomic outcomes:

a) Dollarization and Real Interest Rates

If inflation expectations decline while nominal interest rates remain relatively unchanged in the short term, the real interest rate rises mechanically. Higher real returns on savings can enhance financial intermediation and support capital formation. Nonetheless, the loss of monetary autonomy may limit the central bank's ability to manage cyclical fluctuations in real interest rates. Additionally, by anchoring inflation expectations, dollarization may indirectly contribute to an increase in real interest rates.

b) Dollarization and Country Risk

A credible dollarization program may reduce sovereign risk by signaling commitment to price stability and improving the perceived soundness of the monetary regime. This may lower the country's risk premium and attract foreign capital. However, if the policy is perceived as a substitute for genuine reform or as a desperate stabilization attempt, it may amplify concerns about fiscal sustainability and financial rigidity.

Building on the above theoretical reasoning, this study posits the following testable hypotheses:

H1: Increases in the degree of dollarization are associated with higher real interest rates, and a reduction in the country risk premium.

Beyond the average effects, the stability and credibility of the dollarization process may further shape its dynamic impacts

H2: The magnitude and direction of these effects depend critically on the credibility, timing, and institutional support of the dollarization process. In particular, short-term dynamics may diverge from long-run outcomes.

Collectively, these hypotheses suggest that dollarization could strengthen real financial conditions and reduce sovereign risk, conditional on institutional credibility. And

these hypotheses will be empirically examined in the subsequent sections using both static (OLS) and dynamic (VAR) econometric methods.

IV. Data

The key variables are constructed as follows:

Table 1. : Descriptions of variables

Variable	Description	Measurement	Frequency	Source
DI	Degree of de facto dollarization	$DI = \frac{FCD}{M2}$	Monthly	Fred
R	Real interest rate	$R = I - \pi$	Monthly	BCRA
RISK	Country risk	$Risk = I - I_{USA}$	Monthly	BCRA,World Bank

And Table 2 reports the descriptive statistics for the three key variables over the sample period from June 2010 to December 2024.

Table 2.Descriptive Statistics of Variables

Variable	Mean	sd	Min	Max
DI(%)	43.879	15.863	18.883	91.938
R(%)	-18.557	47.7198	-241.479	8.487
Risk(%)	31.841	24.150	8.023	126.896

The summary statistics indicate substantial variability in both the real interest rate (R) and the country risk premium (Risk). The real interest rate is notably volatile, with values ranging from highly negative real returns (as low as -241.48%) to modestly positive outcomes. This reflects periods of extreme inflationary pressures and macroeconomic instability in Argentina.

In contrast, the dollarization index (DI) exhibits relatively lower volatility, suggesting that shifts in financial dollarization occur gradually over time, fluctuating within a narrower range (approximately 18.88% to 91.94%), and averaging around 43.88% over the sample

period. This suggests that changes in financial dollarization tend to be gradual rather than abrupt.

Figure 1 visually depicts the evolution of these three variables. Following the 2023 presidential election, a sharp rise in real interest rates and a concurrent decline in country risk are observed, while the dollarization index remains largely unchanged. These patterns provide preliminary evidence consistent with the hypotheses that dollarization may stabilize macroeconomic expectations primarily through its impact on financial returns and perceived sovereign risk.

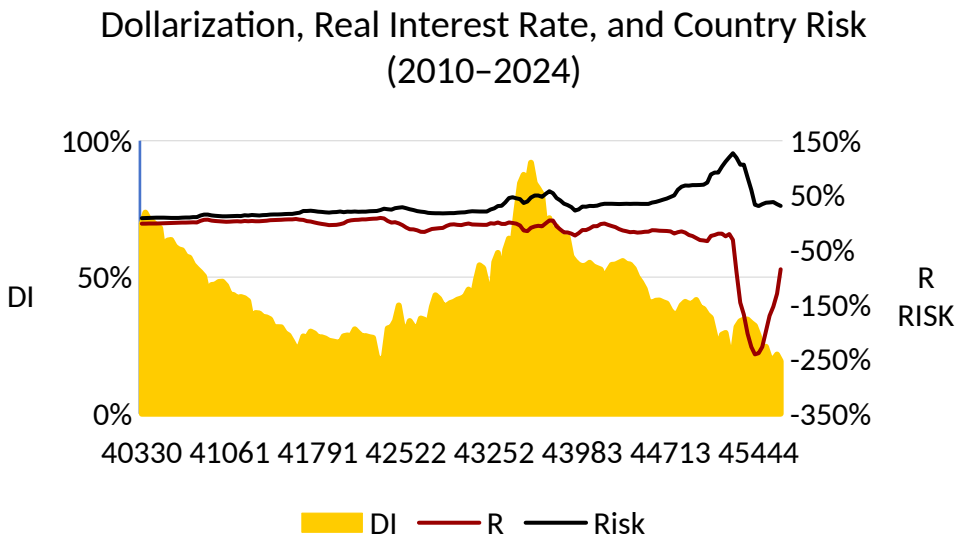


Figure 1. Dollarization, Real Interest Rate, Country Risk (2010-2024)

V. Modeling

a) ARIMA Forecasting Model

This paper began with the ARIMA model, a classic tool in time-series analysis. The ARIMA approach was used to project future trends based on historical data—particularly focusing on inflation rate. The purpose here was not only predictive but also diagnostic: to detect whether any structural changes occurred following the dollarization policy proposal in late 2023.

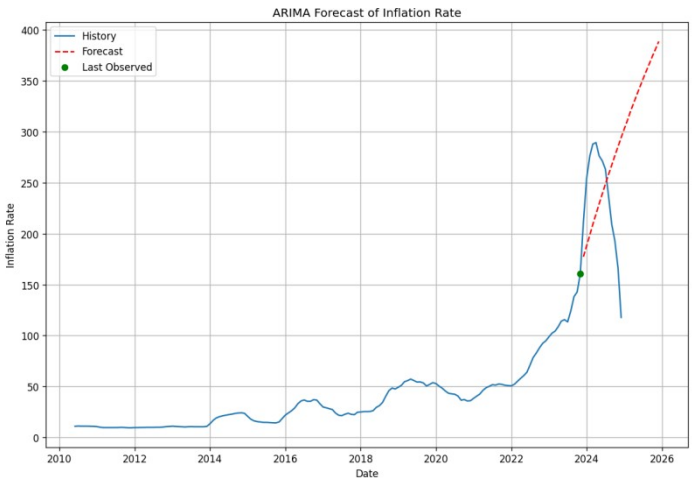


Figure 2: Forecast of Inflation Rate

Figure 2 shows that the comparation between the original data and forecasting data. After 2024, the forecasting inflation continually went higher and did not have a trend to stop or decrease while the real data indicated that after the launch of the new policy in Argentina, there was a sharp falling trend on the inflation rate, much lower than the value forecasted by statistical method. As a result, this paper conclude that Milei's new policy had some positive effect on controlling inflation.

However, while ARIMA can forecast trends, it is fundamentally a univariate model, meaning it doesn't explain causal relationships. Therefore, it is necessary to use more robust tools to capture economic interactions.

b) OLS Regression Model

Next, this paper employed an Ordinary Least Squares (OLS) regression. The core aim here was to analyze the relationship between interest rate (r), investment risk(Risk), and the level of dollarization (DI).

$$R = \alpha + \beta_1 \mathrm{DI} + \beta_2 \mathrm{Risk} + \beta_3 \mathrm{dummy} + \mathrm{e}$$

$$Risk = \alpha + \beta_4 \mathrm{DI} + \beta_5 \mathrm{R} + \beta_6 \mathrm{dummy} + \mathrm{e}$$

To capture the structural policy shift, this paper has introduced a dummy variable, which takes the value of 0 before the dollarization policy was announced on December 1st, 2023, and 1 afterward. This binary variable allows us to isolate the policy's effect on the dependent variables.

The regression result are shown in Chart 1 and Chart 2 which indicated that although both interest rates and risk changed significantly post-policy, these changes couldn't be robustly explained by an increase in the dollarization index. This suggests that while the policy announcement altered market behavior, it didn't materially shift actual economic fundamentals like the degree of dollarization.

Source	SS	df	MS	Number of obs	=	175
				F(3, 171)	=	522.86
Model	35.7268277	3	11.9089426	Prob > F	=	0.0000
Residual	3.89478836	171	.02277654	R-squared	=	0.9017
				Adj R-squared	=	0.9000
Total	39.6216161	174	.227710437	Root MSE	=	.15092

r	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
DI	-.0747648	.0753285	-0.99	0.322	-.2234583	.0739288
Risk	-.0031496	.0004871	-6.47	0.000	-.0041111	-.0021882
dummy	-1.708251	.0481697	-35.46	0.000	-1.803335	-1.613168
_cons	.0646607	.0394271	1.64	0.103	-.0131659	.1424872

Chart 1

Source	SS	df	MS	Number of obs	=	175
				F(3, 171)	=	17.99
Model	24343.9868	3	8114.66227	Prob > F	=	0.0000
Residual	77138.8177	171	451.104197	R-squared	=	0.2399
				Adj R-squared	=	0.2265
Total	101482.805	174	583.234509	Root MSE	=	21.239

Risk	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
DI	-9.838454	10.60502	-0.93	0.355	-30.77205	11.09515
r	-62.38048	9.647084	-6.47	0.000	-81.42319	-43.33777
dummy	-89.96447	18.3469	-4.90	0.000	-126.18	-53.74891
_cons	30.75104	5.073672	6.06	0.000	20.73595	40.76614

Chart 2

c) Vector Autoregression (VAR) Model

To dig deeper into the dynamic relationships among our variables, we adopted a Vector Autoregression (VAR) model. In this framework, we analyzed first-order differences—that is, the changes rather than levels—of country risk (dRisk), interest rate (dr), and the dollarization index (dDI). This approach ensures stationarity, a key requirement for reliable

inference in time-series analysis. Although the Akaike Information Criterion (AIC) indicated that a lag of 4 would be optimal, we extended our analysis to 7 lags to test for long-term interactions. This allowed us to see whether the effects of dollarization policy might be delayed or have persistent impacts over time. Chart 3 and Chart 4 shows the results of the VAR model, which shows that DI has no significant effect on neither risk nor real interest rate, no matter what lag are set.

	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
dRisk						
dRisk						
L1.	.5079393	.1064788	4.77	0.000	.2992447	.7166339
L2.	.1119874	.1304244	0.86	0.391	-.1436398	.3676146
L3.	-.2852213	.1455783	-1.96	0.050	-.5705496	.0001069
L4.	.0822889	.1602478	0.51	0.608	-.231791	.3963689
L5.	-.2326994	.1544914	-1.51	0.132	-.535497	.0700982
L6.	.6215422	.1493384	4.16	0.000	.3288442	.9142401
L7.	-.2268371	.1414432	-1.60	0.109	-.5040607	.0503865
dr						
L1.	-4.393544	8.292849	-0.53	0.596	-20.64723	11.86014
L2.	-5.170982	12.69523	-0.41	0.684	-30.05318	19.71121
L3.	41.3384	14.4338	2.86	0.004	13.04867	69.62813
L4.	-11.74754	15.02809	-0.78	0.434	-41.20206	17.70698
L5.	30.43379	14.10477	2.16	0.031	2.788953	58.07863
L6.	-62.09542	13.05792	-4.76	0.000	-87.68848	-36.50236
L7.	22.13833	10.11058	2.19	0.029	2.32196	41.95469
dDI						
L1.	-5.608752	6.467901	-0.87	0.386	-18.2856	7.0681
L2.	-3.914939	6.379226	-0.61	0.539	-16.41799	8.588114
L3.	9.287239	6.425669	1.45	0.148	-3.30684	21.88132
L4.	8.264608	6.612379	1.25	0.211	-4.695417	21.22463
L5.	-1.564692	6.515117	-0.24	0.810	-14.33409	11.2047
L6.	5.97316	6.463299	0.92	0.355	-6.694673	18.64099
L7.	1.589035	6.46227	0.25	0.806	-11.07678	14.25485
_cons	.1431191	.2643232	0.54	0.588	-.3749448	.661183

Chart 3

dr						
dRisk						
L1.	-.0074363	.0011952	-6.22	0.000	-.0097789	-.0050937
L2.	.0081079	.001464	5.54	0.000	.0052385	.0109773
L3.	-.0094718	.0016341	-5.80	0.000	-.0126746	-.0062691
L4.	-.0018731	.0017988	-1.04	0.298	-.0053986	.0016524
L5.	.0061244	.0017341	3.53	0.000	.0027255	.0095232
L6.	-.0023454	.0016763	-1.40	0.162	-.0056309	.0009401
L7.	-.0042483	.0015877	-2.68	0.007	-.00736	-.0011365
dr						
L1.	1.165618	.093086	12.52	0.000	.9831729	1.348063
L2.	-.8533284	.142502	-5.99	0.000	-1.132627	-.5740295
L3.	.7740868	.1620172	4.78	0.000	.4565388	1.091635
L4.	-.0290489	.168688	-0.17	0.863	-.3596713	.3015736
L5.	-.2746876	.1583239	-1.73	0.083	-.5849967	.0356214
L6.	.1543975	.1465732	1.05	0.292	-.1328806	.4416757
L7.	-.0801841	.1134897	-0.71	0.480	-.3026197	.1422516
dDI						
L1.	.0169041	.0726012	0.23	0.816	-.1253916	.1591998
L2.	-.016993	.0716058	-0.24	0.812	-.1573378	.1233519
L3.	.1049566	.0721271	1.46	0.146	-.03641	.2463232
L4.	.0948221	.0742229	1.28	0.201	-.0506522	.2402964
L5.	.0542997	.0731312	0.74	0.458	-.0890348	.1976342
L6.	.1503395	.0725495	2.07	0.038	.008145	.292534
L7.	.1018435	.072538	1.40	0.160	-.0403284	.2440153
_cons	.0039379	.002967	1.33	0.184	-.0018772	.0097531

Chart 4

VI. Results and Interpretation

From these models, one conclusion stands out: Argentina's risk and interest rate metrics experienced significant shifts after the new policy announcement. However, the level of dollarization itself did not. In other words, the market responded to the announcement emotionally and behaviorally, not structurally. This raises a critical point. The dollarization policy did not directly increase the actual use of the U.S. dollar in Argentina's economy, but it did seem to influence investor perception and short-term expectations.

VII. Conclusion

Based on these findings, we must evaluate the policy as at least partially invalid in its current form. While it may have influenced sentiment, it failed to change the fundamental financial architecture of Argentina's economy.

The policy landscape is also evolving. The recent news shows that on April 11, the IMF, World Bank, and Inter-American Development Bank pledged over $30 billion in aid to Argentina. But this came with strict conditions: abandoning exchange rate controls and prohibiting central bank interventions in the forex market.This means that Argentina's exchange rate is now floating, ranging between 1,000 and 1,400 pesos per USD, a significant departure from past policies of fixed or managed exchange rates. This shift introduces volatility but is aligned with liberalization principles. At that time, the dollarization policy that issued by Milei in Dec.2023 reached to the end.

In conclusion, while Argentina's dollarization policy sparked immediate financial reactions, it fell short of triggering real structural change. Without institutional support and credible enforcement, dollarization remains an ambitious idea rather than a transformative solution.